

Midbrain somatostatin-expressing cells control pain-suppression during defensive states

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The study is a timely and **important** contribution to our knowledge of the circuit mechanisms of fear analgesia. The novel cue-induced analgesia paradigm allowed a **compelling** identification of a brainstem circuit element, i.e., somatostatin-expressing neurons within the ventrolateral periaqueductal grey that project to the rostroventral medulla, in mediating fear analgesia. The vIPAG is a known region of pain modulation, and this study adds key insight to the circuit involved in fear-associated analgesia. This work will be of interest to systems and behavioral neuroscientists, especially those interested in emotional behavior, pain, and/or brainstem function.

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Abstract

Upon threatening situations, animals exhibit a broad range of behavioral and autonomic responses. As such, a crucial adaptive response is the inhibition of pain to facilitate relevant defensive behaviors that promote survival. Whereas the structures and mechanisms involved in fear and pain behavior are well documented, little is known about the precise neuronal mechanisms mediating the emotional regulation of endogenous pain-suppression. Here we used a combination of behavioral, anatomical, optogenetic, and electrophysiological approaches to investigate the role of that somatostatin-expressing cells in the ventrolateral periaqueductal gray matter (SST⁺ vIPAG cells) in the control of analgesia induced during defensive states. Our data indicate that the optogenetic inhibition of SST⁺ vIPAG cells promotes analgesia irrespective of animal's defensive state. In contrast, the optogenetic activation of long-range SST⁺ vIPAG cells that project to the rostral ventromedial medulla (RVM) abolishes the analgesia mediated by fear behavior. Together these results identify a

novel circuit mechanism composed of long-range SST⁺ vIPAG cells projecting to the RVM that regulate analgesia elicited during defensive states.

Introduction

During threatening situations, the inhibition of pain is an essential adaptive response that can be crucial to promote self-preservation^{1–4}. The inhibition of pain responses to a stimulus that would typically elicit pain is defined as analgesia⁵. In addition, previous studies have described that a fearful event can reduce pain sensitivity, a phenomenon referred to as fear-conditioned analgesia (FCA)⁶. In particular, the pharmacological inactivation of the basolateral amygdala abolishes such fear-modulated analgesia, and is associated with an increase in the immediate-early gene c-Fos expression in the central amygdala and periaqueductal gray matter (PAG)⁷. The PAG is a midbrain structure receiving many cortical and subcortical inputs from brain structures involved in fear processing^{8–11}, projecting to the spinal cord primarily through a brainstem relay in the rostral ventromedial medulla (RVM)¹², a key structure involved in nociceptive transmission^{13,14}. Indeed, the ventrolateral part of the PAG (vIPAG) is crucial for the descending control of pain^{13,25} in the dorsal horn of the spinal cord (DH)¹², and several reports indicate that electrical stimulation of the vIPAG selectively inhibits responses to noxious stimuli in a variety of pain test conditions¹². Moreover, analgesia induced upon vIPAG stimulation is opioid-dependent^{15–17} and involves direct descending projections to the RVM and the DH^{18–20,26}. Furthermore, previous research indicate that vIPAG activation induces analgesia through the recruitment of local GABAergic cells²¹, which is consistent with our knowledge of the role of the GABAergic system in the descending control of pain^{22,23}. Thus, the vIPAG is ideally located to allow the emotional regulation of pain behavior, yet the underlying neuronal circuits and mechanisms are largely unknown. Here we investigated the precise neural circuits within the vIPAG that mediates the suppression of pain behavior during threatening situations. We developed a unique behavioral paradigm where a cued-elicited defensive state promotes changes in pain sensitivity. Combining behavioral, anatomical, optogenetic, and electrophysiological approaches, we identified a novel midbrain to brainstem circuit mechanism composed of somatostatin positive (SST⁺) vIPAG cells projecting to the RVM that is critically involved in the suppression of pain behavior during conditioned fear states.

Results

Cued-fear conditioning elicits analgesia

To evaluate the contribution of specific vIPAG cells populations in the fear modulation of pain behavior, we developed a novel cued fear-conditioned analgesia (CFCA) paradigm, consisting of a pain sensitivity assay during the presentation of threat-predicting cues (**Figure 1a**). In this paradigm, an initially neutral stimulus (the conditioned stimulus, CS⁺) becomes associated with a footshock (the unconditioned stimulus, US), whereas another neutral stimulus serves as a control (CS⁻). After conditioning, re-exposure to the CS⁺ elicited freezing behavior, a behavioral readout of associative learned fear³⁷ (**Figure 1b** and Supplementary Figure 1a). Following the fear retrieval session, mice were submitted to the pain sensitivity assay. Mice were placed on a Hot-Plate (HP) apparatus in which the floor plate temperature progressively increases while either CS⁻ or CS⁺ were concomitantly presented. Plate temperature increased, and CSs were continuously played until mice displayed a nociceptive (NC) response reflecting pain behavior (**Figure 1c**), defined by either jumping or hindpaw licking (**Figure 1d**). The detection of a NC response terminated the test session (see **Methods**).

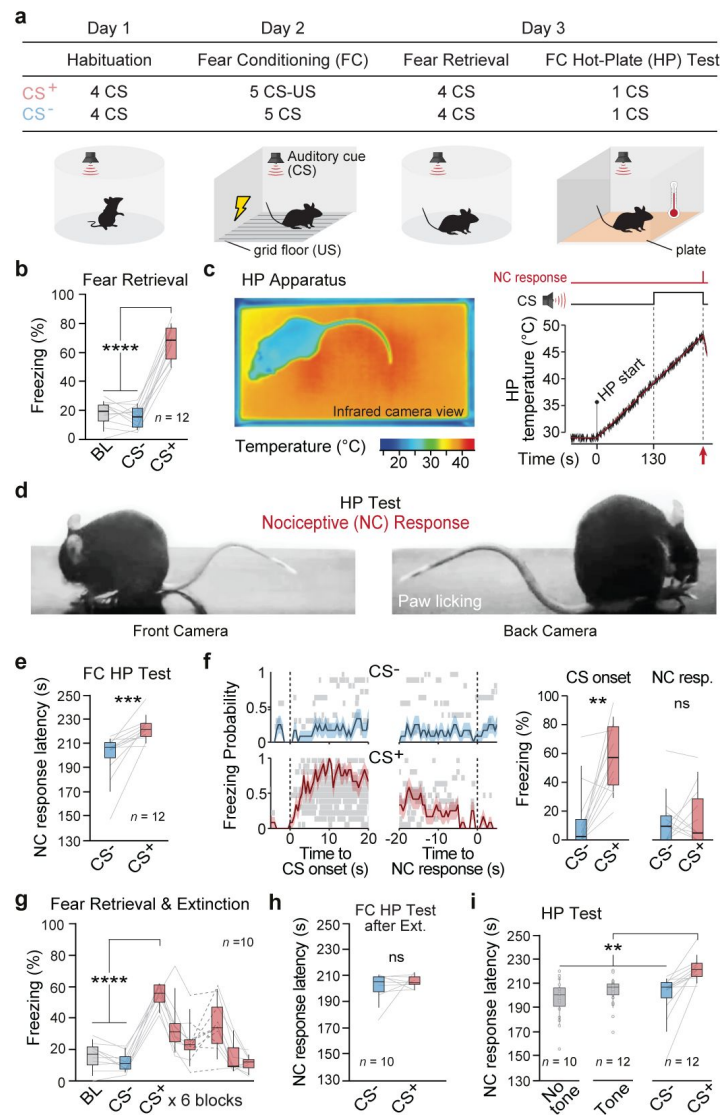


Figure 1.

Fear modulation of pain behavior.

a. Schematic of the setup and the CFCA paradigm. The paradigm consists of a discriminative auditory fear conditioning test followed by a pain sensitivity assay (for details, see Methods). **b.** Average freezing values for CS⁺ was higher than CS⁻ or baseline (BL) periods during retrieval (****, $P < 0.0001$, one-way RM ANOVA, $n = 12$ mice). **c.** Example of the infrared digital thermographic camera view of a mouse during the FC HP test. After 1 min context acclimatization, plate temperature steadily increases at 6 °C per min (HP start). CSs started 130 s after the HP start and continued until mice displayed an NC response (licking the hindpaw or jumping). **d.** Example of assessed hindpaw licking NC response. **e.** Latency of NC response on the FC HP test increased during CS⁺ compared to CS⁻ (***, $P = 0.0005$, Wilcoxon matched-pairs signed rank test, $n = 12$ mice). **f. Left:** dynamics of freezing probability at CS onset and NC response for both CS⁻ and CS⁺ (freezing periods for each individual mouse are displayed by rows in grey; error bars display S.E.M; bin size = 0.5 s). **Right:** average freezing values during 10 s after CS onset and before NC response for CS⁺ were higher than CS⁻ only at CS onset (***, $P_{\text{onset}} = 0.0015$, paired t-test, $n = 12$ mice). **g.** Mean freezing values during fear extinction, 24 CS⁺ were presented across two separate extinction sessions (joined by dashed lines). Mice acquired the CS⁺-US association (1st CS⁺ block vs. BL/CS⁻ ****, $P < 0.0001$, Friedman test, $n = 10$ mice), followed by a rapid extinction (5th & 6th block of CS⁺ vs. BL/CS⁻-ns, $P > 0.05$, one-way RM ANOVA). **h.** After extinction, there was no difference in the NC response latency for both CSs (ns, $P > 0.05$, paired t-test $n = 10$ mice). **i.** Average NC response latencies for the different HP tests. No tone - mice submitted to the HP test without conditioning nor tone presentation. Tone - mice submitted to the HP test paired with unconditioned tone presentation. CS⁻/CS⁺-mice were submitted to the CFCA protocol. (No tone/Tone/CS⁻ vs. CS⁺ **, $P < 0.0001$, Kruskal-Wallis test). For exact P values and test statistics on this and all subsequent figures, see Supplementary Table 1.

We observed a significant delay in the NC response performance for CS⁺ compared to CS⁻ presentations, reflecting that pain-suppression was selectively promoted by the threat-predicting CS (**Figure 1e** and Supplementary Figure 1b). This increased NC latency during CS⁺ presentation could alternatively be explained by a competition between freezing and pain-related responses. However, freezing probability at the time of the NC response for CS⁺ was remarkably low and comparable to CS⁻ levels, suggesting that CS⁺ increased NC response latency was not due to competing freezing responses (**Figure 1f**, see **Methods**). Instead, these results suggest that fear induced by CS⁺ presentations drives the increase in NC latency. Importantly, such states induced by our CFCA paradigm reflected fear associative processes (Supplementary Figure 2). To demonstrate a causal link between cued-fear and analgesia, mice were submitted to an extinction procedure to devalue the CS⁺-US contingency and abolish the negative emotional state during CS⁺ presentations (**Figure 1g**). In these conditions, CS⁺ presentation after extinction had no effects on NC response latency (**Figure 1h**). Furthermore, the absence of auditory cues or the presence of auditory cues not associated with a threat had comparable NC response latencies as those observed during CS⁻ presentations (**Figure 1i**). Finally, previous studies indicate that fear can lead to vasoconstriction, which could result in a redirection of the blood flow to the skeletal musculature, decreasing the temperature in the extremities of the body²⁴. However, monitoring the back and tail temperature of mice submitted to the HP test did not reveal any difference between the CSs presentations, suggesting that increased latency was not due to fear-induced vasoconstriction (Supplementary Figure 1c-e). Overall, our results indicate that the presence of a conditioned threat-predicting cue can selectively promote analgesia.

Photostimulation of SST⁺ vIPAG cells alters pain and fear behavior

To identify the cell types in the vIPAG mediating the analgesic effect in the CFCA task, we focused on GABAergic interneurons, which activity is suggested to be inhibited during analgesia through μ -opioid receptor-dependent mechanisms^{12,27}. Using single-molecule fluorescent in situ hybridization (smFISH, see **Methods**), we confirmed that the majority of vIPAG SST mRNAs were expressed in inhibitory neurons (Supplementary Figure 3). As such, since somatostatin-expressing interneurons (SST) are the most abundant interneurons class in the vIPAG²⁸, we focused on this cell population. Thus, to evaluate the contribution of SST⁺ vIPAG cells in CFCA, SST-Cre mice were injected in the vIPAG with a Cre-dependent AAV expressing the Channelrhodopsin (ChR2), Archaelrhodopsin (ArchT), or GFP while optic fibers were bilaterally implanted targeting this area (**Figure 2a**). Next, mice were submitted to the CFCA task while photostimulating SST⁺ cells in the vIPAG (**Figure 2b**; see **Methods**). After fear retrieval (**Figure 2c**, e; Supplementary Figure 4a-d), the optogenetic activation of SST⁺ vIPAG cells during the HP test had no effect on CS⁻ presentation, whereas the same manipulation performed during CS⁺ blocked the increase in NC response latency compared to GFP controls (**Figure 2d** and Supplementary Figure 5a). In contrast, optogenetic inhibition of SST⁺ vIPAG cells increased NC response latency for both CS⁻ and CS⁺ conditions compared to GFP controls (**Figure 2f** and Supplementary Figure 5b). Importantly, these changes in NC response latencies were not due to motor or aversive effects of the optogenetic manipulations, since photoactivation of SST⁺ vIPAG cells did not induce changes in distance travelled in an open field nor place aversion in a real-time place preference test (Supplementary Figure 5c-f). In addition, we did not observe changes in fear learning or expression when the footshock was replaced or paired with the SST⁺ vIPAG cells photoactivation during the fear conditioning procedure (Supplementary Figure 6).

Altogether, these experiments confirmed that SST⁺ vIPAG activation did not promote pain responses and instead suggested that the SST⁺ vIPAG cells are key elements of an endogenous pain-suppression circuit. Additionally, the optogenetic effect observed during CFCA was not due to alteration of somatostatin levels in our homozygous SST-Cre mice²⁹, as we observed the same effect in heterozygous SST-Cre mice (Supplementary Figure 7a-d). Finally, in contrast with STT⁺

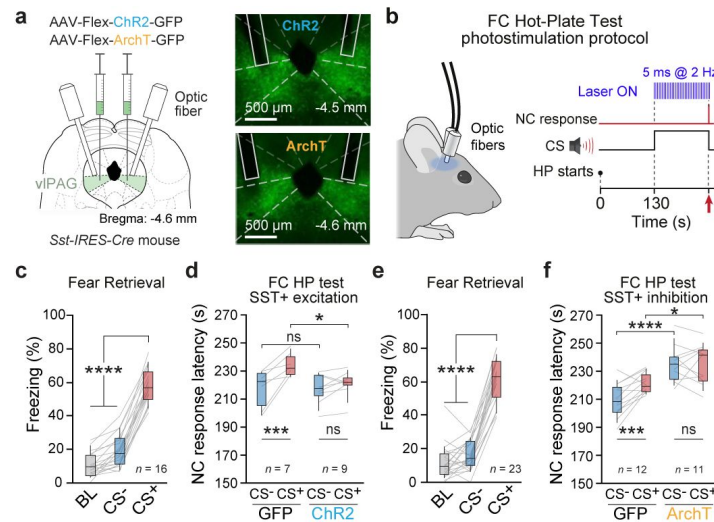


Figure 2.

SST⁺ cells in vIPAG mediate the fear modulation of pain behavior.

a. Sst-IRES-Cre mice were bilaterally injected with viruses that expressed ChR2 (right, upper), ArchT (right, bottom), or GFP (control) in SST⁺ vIPAG cells. **b.** Schematics of the stimulation protocol used for photoactivation of the SST⁺ vIPAG cells: blue light delivered at 2 Hz with 5 ms pulse duration during CSs presentation until NC response. For photoinhibition experiments, green light was delivered continuously. **c, e.** Average freezing values during retrieval for ChR2-(**c**) and ArchT-infected mice (**e**) and their respective GFP controls. The opsin and respective control groups were pooled together because no difference was found in the conditioning level (see Supplementary Figure 4). Average freezing values during CS⁺ was higher than CS⁻ or baseline (BL) periods (****, $P < 0.001$, one-way RM ANOVA, (**c**) $n = 16$ mice and (**e**) $n = 23$ mice). **d.** Photoactivation of SST⁺ vIPAG cells abolished the analgesic effect induced by fear (*, $P = 0.0112$, opsin x CS - two-way RM ANOVA, $n = 7$ GFP, $n = 9$ ChR2). The NC response latency for the CS⁺ was significantly different between the ChR2 and GFP groups (*, $P = 0.0344$, Bonferroni post-hoc test). For the ChR2 group, the NC response latency during CS⁺ was equivalent to the CS⁻ (ns, $P = 0.3876$, Bonferroni post-hoc test). On the contrary, the NC response latency between the CSs differed for the GFP group (***, $P = 0.0003$, Bonferroni post-hoc test). **f.** Photoinhibition of SST⁺ vIPAG cells increased the analgesic effect for the ArchT group when compared to the GFP (*, $P = 0.0037$, opsin x CS - two-way RM ANOVA, $n = 12$ GFP, $n = 11$ ArchT). The NC response latency for the CS⁻ and CS⁺ was significantly different between the ArchT and GFP group (CS⁻: ****, $P < 0.0001$, Bonferroni post-hoc test; CS⁺: *, $P = 0.0265$, Bonferroni post-hoc test). For the GFP group, the latency of NC response was higher for the CS⁺ trials when compared to the CS⁻ trials (***, $P = 0.0003$, Bonferroni post-hoc test), yet this was not the case for the ArchT group (ns, $P > 0.999$, Bonferroni post-hoc test).

vIPAG cells, the optogenetic inhibition of another important class of vIPAG inhibitory cells expressing the vasoactive intestinal peptide (VIP⁺) did not change the NC response latencies during CS⁻ or CS⁺ presentations (Supplementary Figure 7e-h).

Because our CFCA paradigm is dependent on fear associative processes (Figure 1 and Supplementary Figure 1-2), it is possible that the optogenetic manipulation of SST⁺ vIPAG cells may have altered the expression of fear behavior. To control for this possibility, we optogenetically activated or inhibited SST⁺ vIPAG cells during a fear retrieval session following auditory fear conditioning (Figure 3a, b). Our data indicate that whereas the optogenetic inhibition of SST⁺ vIPAG cells had no effect on freezing behavior relative to GFP controls, their optogenetic activation reduced freezing levels (Figure 3c, d). Altogether, our data show that the local SST⁺ vIPAG network can modulate both fear expression and the analgesia promoted during defensive states.

Photostimulation of SST⁺ vIPAG cells regulates spinal cord nociceptive transmission

Consistent with our observations, previous studies found that the vIPAG is involved in fear-conditioned analgesia⁷. It is conceivable that the SST⁺ vIPAG cells are indeed part of a circuit involved in the fear regulation of endogenous pain-suppression. To evaluate the effect of the manipulation of SST⁺ vIPAG cells on analgesia independently of the expression of fear behavior, we evaluated the impact of SST⁺ vIPAG cells manipulation on the control of nociceptive transmission directly at the level of the spinal cord DH. Therefore, we performed extracellular local field potential and single-unit recordings in the spinal cord DH during optogenetic stimulation of SST⁺ vIPAG cells in anesthetized mice (Figure 4a). Consistent with previous studies³⁰, nociceptive stimulation induces early and late neuronal responses at the level of the spinal cord DH (A-comprising non-nociceptive A β and nociceptive Ao and C-fiber mediated, respectively; Figure 4a). While optogenetic activation of SST⁺ vIPAG cells during paw suprathreshold electrical stimulation increased the magnitude of late nociceptive field potentials, the optogenetic inhibition reduced it (Figure 4b, c). To determine if this effect was specific to the nociceptive network, we recorded wide dynamic range (WDR) cells in the DH, which receive both tactile (A β) and nociceptive information (Ao and C)³⁰. We observed that SST⁺ vIPAG cells photoactivation increases both fast nociceptive (Ao-fiber) and slow nociceptive (C-fiber) responses without affecting A β (Figure 4d). Moreover, the optogenetic inhibition of SST⁺ vIPAG cells specifically inhibited nociceptive responses while A β responses remained unaffected (Figure 4e). In addition, we performed subliminal electrical stimulations that did not elicit nociceptive neuronal responses. In this condition, we observed that the removal of analgesia by the activation of SST⁺ vIPAG cells leads to WDR responses at latencies corresponding to nociceptive Ao and C-fibers (Figure 4f). Finally, the optogenetic activation of SST⁺ vIPAG cells had no effect on purely tactile fast latency spinal neurons (Supplementary Figure 8). Consistent with our optogenetic manipulations during the CFCA task (Figure 2), these data suggest that under analgesia promoted by isoflurane, activation and inhibition of SST⁺ vIPAG cells increased or decreased nociceptive transmission by specifically promoting pronociceptive and antinociceptive responses in the spinal cord DH, respectively.

SST⁺ vIPAG-RVM-spinal cord pathway activation abolishes analgesia induced during defensive states

We next evaluated whether SST⁺ vIPAG cells mediate their analgesia modulation by contacting WDR cells directly in the spinal cord or through the rostro ventromedial medulla (RVM), a structure known to receive direct projections from vIPAG and mediate pain modulation^{18,19} (Figure 5a). Anatomical tracing revealed large labelling of SST⁺ fibers in RVM (Figure 5b) and only sparse labelling within the spinal cord (SC; data not shown). By combining fluorogold

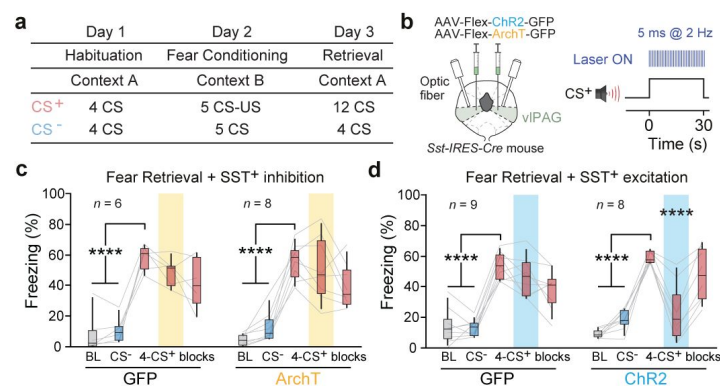


Figure 3.

Activation of SST⁺ vPAG cells reduced fear expression

a. Protocol for optogenetic manipulation during fear retrieval. Day 1 and 2 were done as described previously for the CFCA paradigm. During retrieval, there were 12 CS⁺ presentations divided into three blocks. **b.** The stimulation protocol used for photoactivation of the SST⁺ vPAG cells. The optogenetic manipulation was done during the 2nd block of the CS⁺ presentation (analogously to CFCA manipulations, **Figure 2b**). **c.** Light inhibition of SST⁺ vPAG cells did not modulate freezing levels (ArchT ns, $P = 0.9396$, opsin x CS two-way RM ANOVA, $n = 6$, GFP, $n = 8$). **d.** Photoactivation of the SST⁺ vPAG cells had no effect on the GFP group, but it transiently decreased the freezing levels for the ChR2 group (***, $P < 0.0001$, opsin x CS two-way RM ANOVA, $n = 9$ GFP, $n = 8$ ChR2).

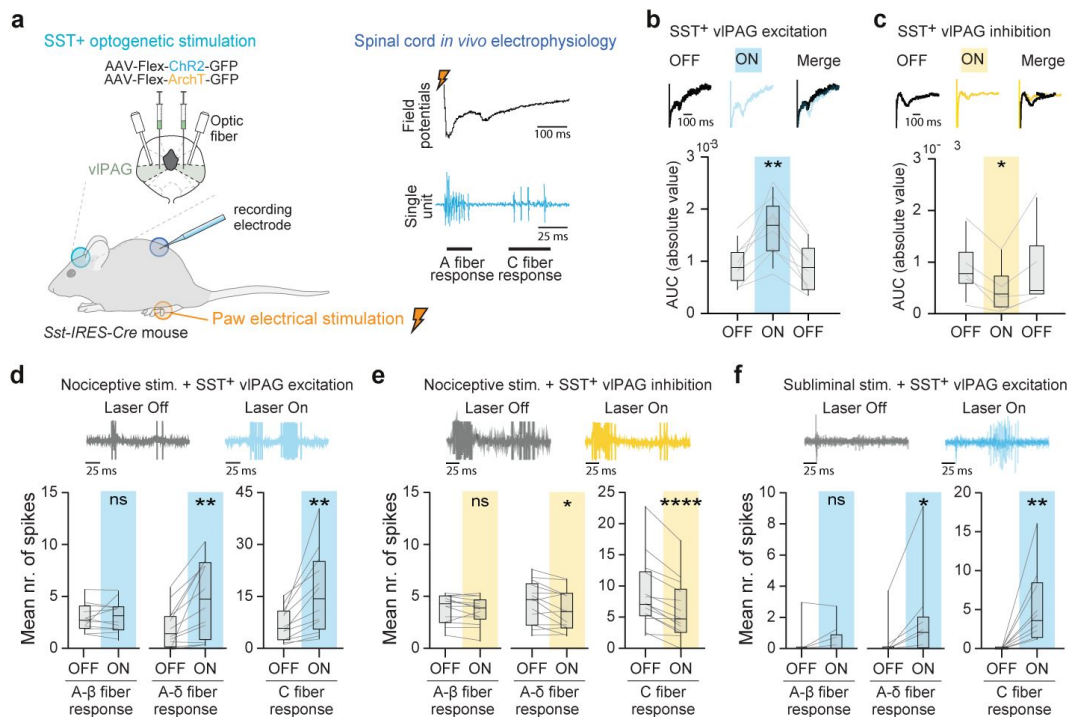


Figure 4.

Manipulation of SST⁺ vPAG cells alters spinal cord-related pain signals.

a. Schematics of *in vivo* anesthetized experiments. Extracellular local field potential and single-unit recordings in the spinal cord DH were performed during optogenetic stimulation of SST⁺ vPAG cells with concomitant noxious electrical paw stimulation. **b.** Example of average nociceptive field potentials in the lumbar SC before (OFF), during (ON), and after (OFF) photoactivation of SST⁺ vPAG cells (top). Photoactivation induces a significant increase in the nociceptive fields (bottom; **, $P = 0.0001$, one-way RM ANOVA, $n = 8$ mice). **c.** Analogous to panel b, but for photoinhibition of SST⁺ vPAG cells. Photoinhibition induces a significant decrease in the nociceptive fields (bottom; *, $P = 0.0275$, one-way RM ANOVA, $n = 5$ mice). **d.** Example of WDR single-unit activity before and during photoactivation of SST⁺ vPAG cells (top). Photoactivation of SST⁺ vPAG cells induces a significant and global increase in WDR response to both A δ - and C-mediated nociceptive fibers (bottom; **, $P_{A\delta \text{ fiber}} = 0.0022$, paired t-test, $n = 11$ cells; **, $P_{C \text{ fiber}} < 0.0033$, paired t-test, $n = 11$ cells). **e.** Analogous to panel d, but for photoinhibition of SST⁺ vPAG cells. Photoinhibition induces a significant and specific inhibition of WDR response to A δ - and C-fibers (bottom; *, $P_{A\delta \text{ fiber}} = 0.0241$, ****, $P_{C \text{ fiber}} < 0.0001$, $n = 14$ cells, Wilcoxon matched-pairs signed rank test). **f.** Example traces of single-unit recordings of WDR cells with subthreshold electrical stimulation accompanied by photoactivation of SST⁺ vPAG cells (top). The photoactivation of SST⁺ vPAG cells elicits WDR response to A- and C-mediated peripheral fibers (bottom; *, $P_{A\delta \text{ fiber}} = 0.0156$, $n = 10$ cells; **, $P_{C \text{ fiber}} = 0.0020$, $n = 10$ cells, Wilcoxon matched-pairs signed rank test).

retro labelling from the SC and GFP anterograde labelling from SST⁺ vIPAG cells (**Figure 5c**), we observed RVM cells projecting to the DH with close apposition of SST⁺ putative boutons (**Figure 5d**), suggesting that SST⁺ vIPAG cells project to the RVM and contact DH-projecting RVM cells. Optogenetic activation of SST descending fibers above the SC has no effect on WDR neural activity upon paw electrical stimulation, suggesting that SST⁺ vIPAG long-range projections to SC did not play a major role in endogenous pain-suppression (**Figure 5e**). In contrast, photoactivation of SST⁺ vIPAG inputs to the RVM increased both nociceptive (Ao and C-fiber) responses (**Figure 5f**, Supplementary Figure 9). Altogether, these data revealed that the SST⁺ vIPAG-RVM-DH pathway mediates endogenous pain-suppression.

We next addressed if the SST⁺ vIPAG-RVM-DH pathway was specific in the mediation of endogenous pain-suppression induced by defensive states. Thus, SST-Cre mice were injected in the vIPAG with a Cre-dependent AAV expressing the ChR2 or GFP, and a single optic fiber was implanted above the RVM (**Figure 6a**). While photoactivation of the SST⁺ vIPAG-RVM pathway promoted nociception (**Figure 5f**), it did not impact fear expression as freezing levels remained unchanged (**Figure 6b**). Consistently, activation of the SST⁺ vIPAG-RVM pathway during CFCA had no impact on CS⁻ presentation, whereas the same manipulation performed during CS⁺ blocked the increase in NC response latency compared to GFP controls (**Figure 6c, d**). Overall, these data suggest the existence of two populations of SST⁺ vIPAG cells mediating fear expression and analgesia induced by defensive states. (**Figure 6e**). Altogether, our results indicate that the SST⁺ vIPAG-RVM-DH pathway is critically involved in the analgesia induced during defensive states.

Discussion

In this study, we established that SST⁺ vIPAG cells control the analgesia during defensive states. Endogenous pain-suppression was mediated by long-range SST⁺ vIPAG cells projecting to the RVM, which in turn transmitted the pain signal to the DH of the spinal cord (**Figures 2** and **5**). Finally, our data indicated that SST⁺ vIPAG cells can regulate both fear expression and the analgesia promoted during defensive states, and that the expression of these two behaviors can be segregated by precisely targeting the SST⁺ vIPAG-RVM pathway (**Figure 6**). Together these results identify a novel midbrain to brainstem circuit mechanism composed of long-range SST⁺ vIPAG cells projecting to the RVM that regulate analgesia elicited during defensive states.

Our observations of the involvement of vIPAG in our cued-based fear conditioned analgesia paradigm are consistent with previous studies in contextual-based FCA involving this region⁷. In such FCA studies⁶, animals undergo contextual fear conditioning followed by pain behavior testing in the same conditioning context, limiting a temporally precise elicitation of aversive states. Instead, our behavioral paradigm goes beyond these limitations by promoting defensive states through a cued conditioned stimulus, advancing toward temporal-specific manipulations such as optogenetics, and allowing us to study the phenomena with a within-subject experimental design by subsequent CS^{+/−} presentations. Nevertheless, since the circuits involved in contextual and cued fear may differ^{31,32}, this might also imply different circuits involved in the emotional modulation of pain-suppression behavior. To our knowledge, this is the first demonstration of a cued fear-conditioned analgesia model in rodents. Addressing possible differences in the circuitry involved in contextual and cued fear-induced analgesia will require further investigation.

Our observation that the photoactivation of SST⁺ vIPAG cells not only abolished the analgesia during defensive states but also decreased freezing expression has far-reaching functional consequences. Two independent previous studies found that long-range inhibitory inputs from the central medial amygdala contact inhibitory cells within the vIPAG, implicated in different roles: the modulation of fear behavior²¹ and nociceptive transmission³³. Notably, both studies describe the involvement of vIPAG inhibitory cells in a disinhibitory circuit mechanism of long-

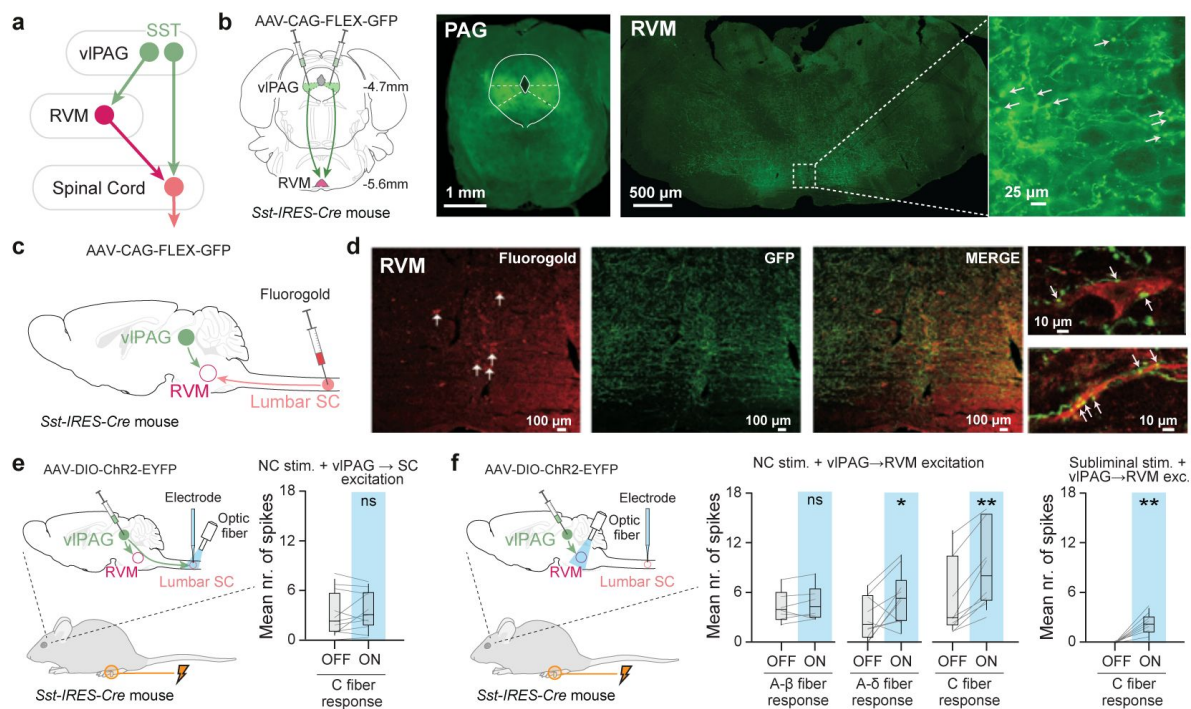


Figure 5.

SST⁺ vIPAG-RVM-DH pathway activation removes analgesia.

a. Schematics of possible SST⁺ vIPAG circuits mediating analgesia by long-range projections directly to the SC or, alternatively, by projecting to the SC via the rostral ventromedial medulla (RVM). **b.** Sst-IRES-Cre mice were injected with an anterograde AAV Cre-dependent GFP virus in the vIPAG (left). Example of SST fibers labelling at the level of the RVM (middle). Higher magnification (right) reveals putative axonic buttons in the RVM (examples indicated by white arrows). **c.** Sst-IRES-Cre mice were injected concomitantly with anterograde AAV Cre-dependent GFP virus in the vIPAG and retrograde fluorogold in the lumbar DH of the SC. **d.** Fluorogold positive cells (red) cross SST⁺ vIPAG fibers in the RVM (green). Higher magnification (right) shows close contacts between the putative SST⁺ button and fluorogold⁺ cells or fibers (white arrows). **e.** Sst-IRES-Cre mice were injected in the vIPAG with AAV Cre-dependent ChR2 virus and optic fiber placed above the lumbar SC. Single-unit recordings of WDR cells while photoactivation of the SST⁺ vIPAG SC fibers and electrically stimulating the paw in anesthetized mice (left). Photoactivation of vIPAG projections to SC did not affect nociceptive transmission (right; ns, $P = 0.3683$, paired t-test, $n = 11$ cells). **f.** Same experimental design as in panel e, except that the optic fiber was placed above the RVM (left). Photoactivation of SST⁺ vIPAG inputs to the RVM induces a significant increase in WDR response to both A δ - and C-mediated nociceptive fibers stimulation (middle; *, $P_{A\delta \text{ fiber}} = 0.0346$, Paired t-test, $n = 10$ cells; **, $P_{C \text{ fiber}} = 0.0020$, Wilcoxon matched-pairs signed rank test, $n = 10$ cells). Removal of analgesia by activation of SST⁺ vIPAG-RVM induces a significant increase in WDR response to C-fibers responses (right; **, $P = 0.0078$, Wilcoxon matched-pairs signed rank test, $n = 9$ cells).

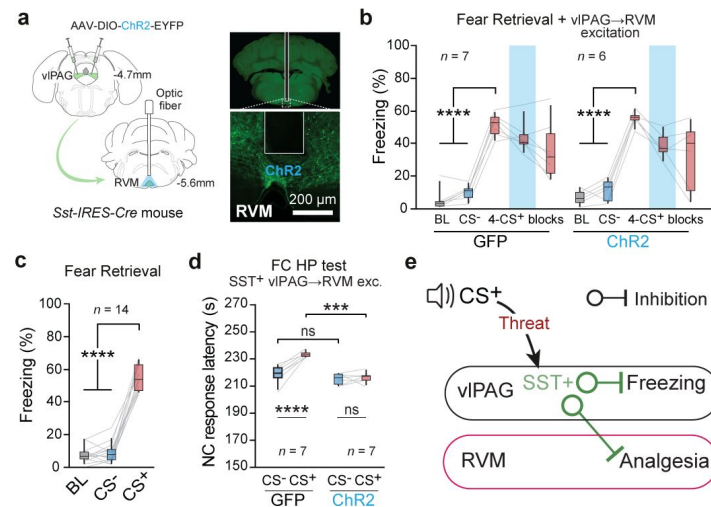


Figure 6.

SST⁺ vPAG-RVM-SC pathway activation abolishes analgesia induced by CFCA.

a. *Sst-IRES-Cre* mice were bilaterally injected with an AAV expressing ChR2 or GFP in the vPAG, and optic fibers were implanted above the RVM (left). Representative example of *Sst*⁺ vPAG terminals in the RVM (right). **b.** Photoactivation of *Sst*⁺ vPAG-RVM pathway did not modulate freezing levels (ns, $P = 0.6443$, opsin $\times$ CS two-way RM ANOVA; CS⁺ 2nd block, GFP vs. ChR2 ns, $P > 0.9999$, Bonferroni post-hoc test, $n = 7$ GFP, $n = 6$ ChR2). **c.** Average freezing values during retrieval for pooled ChR2 and GFP-infected mice (groups were pooled together since no difference was found in the conditioning level). Average freezing values during CS⁺ were higher than CS⁻ or baseline (BL) periods (****, $P < 0.0001$, one-way RM ANOVA, $n = 14$ mice). **d.** Photoactivation of *Sst*⁺ vPAG-RVM pathway abolished the analgesic effect modulated by fear (**, $P = 0.0063$, opsin $\times$ CS - two-way RM ANOVA, $n = 7$ GFP, $n = 7$ ChR2). For the ChR2 group, the NC response latency during CS⁺ was equivalent to the CS⁻ (ns, $P > 0.9999$, Bonferroni post-hoc test). On the contrary, the NC response latency between the CSs differed for the GFP group (****, $P < 0.0001$, Bonferroni post-hoc test). **e.** Working model schematics for *Sst*⁺ control analgesia induced during defensive states. Activation of *Sst*⁺ cells in the vPAG is associated with the inhibition of fear-induced freezing behavior. Selective activation of RVM projecting *Sst*⁺ vPAG cells is associated with an inhibition of analgesia induced during defensive states.

range projecting excitatory cells as the underlying neural substrates responsible for such modulations^{21,33}. We observed that SST⁺ is the most abundant type of inhibitory cells in the vPAG (Supplementary Figure 3). Thus, it is conceivable that the SST⁺ vPAG cells are indeed part of a circuit involved in the emotional regulation of endogenous pain-suppression. Notably, global manipulation of SST⁺ vPAG cells cannot disentangle their role in analgesia and fear behavior at the local level (**Figure 2** and **3**). Because the studies mentioned above exclusively addressed the contribution of vPAG circuits in either pain behavior³³ or fear behavior²¹, these studies could have been oblivious to alternative behavioral readout interpretations. Here we demonstrated that SST⁺ vPAG cells projecting to the RVM are involved in the modulation of analgesia induced during defensive states without impacting the behavioral expression of fear (**Figure 6**). Thus, our results suggest that fear and analgesia signals rely on overlapping neuronal circuits and that the discrimination between these two states could be mediated within the vPAG network. Altogether, our data is consistent with the existence of different populations of SST⁺ cells within the vPAG responsible for the mediation of fear behavior and analgesia information, defined by their output connectivity.

The lateral inhibition model of analgesia within the PAG relies on the activation of PAG excitatory cells projecting to the RVM. More precisely, in this model, analgesia is thought to occur through an opioid-dependent inhibition of GABAergic cells, ultimately disinhibiting excitatory cells projecting to the RVM^{12,19,20}. Our data indicate that in addition to this disinhibitory mechanism, analgesia is also mediated by the inhibition of a direct long-range SST⁺ neurons projection onto RVM cells projecting to the DH (**Figure 6**). Our results align with the parallel inhibition-excitation model, where inhibitory and excitatory cells form two distinct, parallel descending pathways for pain modulation³⁵. Indeed, previous research demonstrated the presence of an inhibitory pathway projecting throughout the PAG–RVM–spinal cord dorsal horn neuraxis³⁶. Our results complement this study by suggesting that this previously proposed inhibitory pathway is mediated by GABAergic SST⁺ vPAG cells and has a functional role in mediating analgesia.

We identified a novel midbrain to brainstem circuit mechanism composed of SST⁺ vPAG cells that project to the RVM, which regulates analgesia during defensive states. Substantial progress has been made that outlined the midbrain circuitry in endogenous pain-suppression pathway^{12,23,37}. However, only more recently, high cortical areas have also been shown to be involved in analgesia^{7,38}. Previous reports describe connections between CeL and vPAG inhibitory neurons in the modulation of pain behavior^{21,34}. Future work is required to elucidate as to whether the exact circuit that regulates analgesia are mediated by different negative emotional states such as fear, depression or stress.

Recent human reports reveal strong comorbidity between mood and anxiety disorders with and pain-related disorders³⁹, suggesting that the deregulation of the circuits mediating the interplay between emotional and pain behavior could be at the core of such psychiatric conditions. Comprehensive knowledge about the circuits mediating the emotional regulation of pain behavior is urgently needed. We identified an SST⁺ cell type- and pathway-specific involvement in the mechanisms that mediate the emotion-dependent regulation of pain behavior. Thus, these findings may empower the development of more effective analgesic therapies by the improvement of further refined pharmacological targets.

Methods

Subject details

We used either male C57BL6/J mice (Janvier), heterozygous or homozygous Sst-IRES-Cre mice (Jackson laboratory), or heterozygous VIP-IRES-Cre mice (Jackson Laboratory) age 8-14 weeks that were individually housed under a 12 h light-dark cycle and provided with food and water ad

libitum. All procedures were performed in accordance with standard ethical guidelines (European Communities Directive 86/60-EEC) and were approved by the committee on Animal Health and Care of Institut National de la Santé et de la Recherche Médicale and the French Ministry of Agriculture and Forestry (agreement #A3312001).

Behavioral apparatus

Cued fear-conditioned analgesia task was performed in three different contexts ([Figure 1a](#)). Context A was used for *Habituation* and *Retrieval* and consisted of a plexiglass cylinder (25 x 24 cm diameter) with a grey, smooth plastic floor and house lights (16 lux). Context B was used for *Fear conditioning* and consisted of a square plexiglass (25 x 40 cm) with a grid floor connected to a shocker (Coulbourn Instruments) and brighter house lights (40 lux). A total of 5 scrambled footshocks of 1 s duration and intensity of 0.8 mA were delivered via the grid floor and served as the unconditioned stimulus (US). Context A and B were cleaned, respectively, with 70% ethanol or 1 % acetic acid between different mice. Both contexts contained an infrared beam detection that automatically scored freezing periods, determined by no change in infrared beam crossing for at least 2 s.

Context C was used for the *FC Hot Plate (HP) test*. A steady increase in temperature was controlled by the Incremental Hot/Cold Plate Analgesia Meter (IITC) device. To restrain the mice's movement, the HP apparatus had a testing surface (plate) enclosed in a square plexiglass surface (20.3 x 10 x 20.5 cm) and house lights (100 lux). The mice's temperature and surroundings were recorded with an infrared digital thermographic camera (Testo 885) placed ~ 50 cm above the testing surface. The thermal camera had a spatial resolution of 320 × 240 pixels, a sampling rate of 25 Hz, and thermal sensitivity of 0.03 °C at 30 °C. The testing surface of context C was cleaned with water between different mice. All three contexts were enclosed in an acoustic foam isolated box with speakers mounted on the top of each compartment. The auditory conditioned stimulus (CS) consists of either 7.5 kHz or white-noise 50 ms pips at 1 Hz repeated 27 times, 2 ms rise and fall, and 80 dB sound pressure level.

Open field task was performed in a square plexiglass arena (36 x 36 x 25cm). A LED mounted on the top-right side of the arena signalled the start and end of different epochs for offline analyses. A video camera recorded from above the arena at 30 fps for offline video-tracking purposes.

Real-Time Place Preference (RTPP) task was performed in a shuttlebox consisting of a plexiglass box (40 x 10 x 30 cm) with a floor grid, where a small plastic hurdle (1 cm height) divided the arena into two equal compartments while infrared beams detection automatically monitored the mice shuttling between compartments (Imetronic). A video camera recorded from above the arena at 30 fps for offline video-tracking purposes.

For both the open field and RTPP task, a free user video-tracking software (idTracker: Tracking individuals in a group by automatic identification of unmarked animals) together with in-house codes in Matlab (The MathWorks, Inc., Natick, MA, USA) were used to analyze each condition.

Behavioral paradigm

Cued fear-conditioned analgesia (CFCA)

C57BL6/J mice (n = 12) were habituated to the context and tones (day 1). Four white-noise (CS⁻) and four 7.5 kHz (CS⁺) tones were presented sequentially and without US reinforcement. In Fear conditioning (day 2), five CS⁻ and five CS⁺ were presented in an intermingled fashion. The CS⁺ presentations were paired with a mild footshock (US) at tone offset, whereas the CS⁻ was never

reinforced. The Fear retrieval session (Day 3) was done 24h after conditioning and in the same context as the habituation session. As in habituation, four CS⁻ and four CS⁺ were presented sequentially and without US reinforcement (**Figure 1a** [↗](#)).

Since the focus of this study was the emotional modulation of pain sensitivity, it was compulsory to evaluate the associative fear levels before measuring its impact on pain sensitivity. Two indices were computed: the discrimination index (DI), to assess the discrimination between CS⁻ and CS⁺, and the conditioning index (CI), which indicated the level of freezing to the tone predicting the US.

These indexes were calculated as follows: $DI = \frac{(Freezing\ to\ CS^+) - (Freezing\ to\ CS^-)}{(Freezing\ to\ CS^+) + (Freezing\ to\ CS^-)}$ and $CI = DI \times (Freezing\ to\ CS^+)$. Based on preliminary data, mice were only submitted to the FC HP test if $DI \geq 0.4$ & $CI \geq 0.3$. Mice that did not fit the criteria were conditioned a second time. There was a minimum interval of 2 h between the retrieval and the FC HP test.

The FC HP test consisted of two trials, one where the CS⁺ was presented and another where the CS⁻ was presented. The two trials were counterbalanced within the group. The testing surface was set at 30 °C. After a 60 s acclimatization period, its temperature gradually increased at 6 °C per minute (HP start). Tone presentation (CS⁺ or CS⁻) started 130 s after the HP start. Temperature increase and tone presentation terminated concomitantly with the display of a nociception response. Valid nociception readout responses included jumping or licking the hindpaw. The effect of the negative emotional modulation on pain sensitivity was assessed by comparing the latency (or temperature) of the NC response between the two HP trials. There was a minimum of 30 min interval between the two trials for each mouse, during which mice returned to their home cage.

Freezing during the HP test was assessed by automatically analyzing thermographic camera videos using custom scripts (available at: github.com/djercog/TestoFreez). Briefly, individual video frames were binarized to define a mask for the mouse within each frame. Consecutive binarized frames were then subtracted, and the number of non-zero pixels on the resultant image determined if the mouse was considered immobile when below a certain threshold (defined to approximate manual freezing estimation on random videos). Immobilization episodes of at least 0.5 s were considered as freezing periods. In contrast, non-immobilization episodes shorter than 0.1 s were merged together with the neighbouring detected freezing episodes.

Extinction training

On days 3 and 4, mice (n = 10) were submitted to an extinction training protocol established by Courtin et al.⁴⁰ [↗](#). Briefly, there were 4 CS⁻ and 12 CS⁺, presented in a non-reinforced manner (**Figure 1g** [↗](#)). Mice were considered as having successfully extinguished the fear expression if the level of freezing in the last 4 CS⁺ was not statistically different from the CS⁻. After fear extinction, mice were submitted to the FC HP test, using the same parameters as in the CFCA paradigm.

Stability training

The first three days consisted of the classical CFCA paradigm. On day 4, mice (n = 10) repeated the protocol applied on day 3 (Supplementary Figure 2e-i). All mice that passed the fear conditioning criteria ($DI \geq 0.4$ & $CI \geq 0.3$) on day 3 were kept for the following day, independent of their level of Conditioning on day 4.

Vasoconstriction assay

To determine if vasoconstriction was a confounding factor in the FC HP test outcomes, we used a modified version of the CFCA paradigm (**Figure 1f** [↗](#)). During the HP test (n = 13), the temperature was not gradually increased but maintained at 30 °C while CSs were presented. The tones (CS⁺ or CS⁻) were presented for 120 s, and mice were kept for an additional 50 s before the HP trial.

terminated. Offline, with the infrared videos, mice's back and tail were measured at 30 s intervals. For each measuring point, the temperature of three spots of the two body part was averaged to create the body temperature for the back and tail for a given time point and mouse.

Basal nociception assay (No tone)

Naïve mice ($n = 10$) were submitted to the HP test (**Figure 1i** and Supplementary Figure 2a-d). Each mouse underwent two identical trials. The HP test was similar to the one described in the CFCA paradigm, except no tones were presented. There was a minimum of 30 min interval between the two trials, during which mice returned to a restful state in their home cage. Since the two trials were not significantly different, for simplification, in panel i from **Figure 1**, the two trials were pooled together.

Tone-specific nociception assay

Naïve mice ($n = 12$) were submitted to the HP test without auditory fear conditioning (**Figure 1i** and Supplementary Figure 2a-d). Each animal underwent two trials, one trial with a 7.5 kHz tone presentation and another with a WN tone presentation. The tone presentation was counterbalanced. The HP test parameters were similar to the one described for the CFCA, except that both tones were unconditioned. Since the two trials were not significantly different, for simplification, in panel i from **Figure 1**, the two trials were pooled together.

Open Field assay

This test was used to determine the effect of optogenetic stimulation on locomotion. Therefore, only mice from optogenetic experiments were submitted to this assay. Mice could freely move during the entire test. The locomotion assay had a total duration of 9 min and was divided into 3 min epochs. The first and third epochs were OFF periods in which no optical stimulation occurred. During the second epoch, mice received photostimulation (ON period). The optogenetic stimulation effect was analyzed by comparing the overall distance travelled between the OFF and ON epochs (Supplementary Figure 5c, d). Mice injected with GFP were used to test the effect of heat and light on the stimulation itself. Mouse tracking data was extracted from video recordings using idTracker⁴¹ and imported to Matlab for further analysis using custom scripts to calculate the total distance travelled during stimulation.

Real-time Place Preference assay

Only mice from optogenetic experiments were submitted to this assay. During the entire duration of the assay, mice could freely shuttle between the two compartments. Under a closed-loop stimulation, mice received photo-stimulation upon entry into one of the two compartments. The stimulated compartment was counterbalanced within the group. The optical stimulation effect was assessed by comparing the time spent in the stimulated compartment between the ChR2 and GFP groups (Supplementary Figure 5e, f). Permanence time in the stimulated compartment was automatically assessed by apparatus software (Imetronic).

Virus injections and optogenetics

For optogenetic manipulation of SST⁺ cells in the vIPAG, 0.15-0.2 μ L of either ChR2 (AAV5-EF1a-DIO-hChR2(H134R)-EYFP, titer: 3.2×10^{12} - Vector Core, University of North Carolina), ArchT (AAV9-CAG-FLEX-ArchT-GFP, titer: 4.7×10^{12} - Vector Core, University of North Carolina) or GFP (AAV5-FLEX-GFP, titer: 4.5×10^{12} - Vector Core, University of North Carolina) were bilaterally injected into the vIPAG of 8/9 weeks old Sst-IRES-Cre mice from glass pipettes (tip diameter 20-30 μ m) at the following coordinates relative to bregma: - 4.4 mm AP; $\pm$ 1.5 mm ML; -2.45 mm DV from dura, with a 20 degrees angle. Injection coordinates for manipulating VIP⁺ vIPAG cells were the following: -4.4 mm AP; $\pm$ 1.35 mm ML; -2.5 mm DV from dura, with a 20 degrees angle.

At two weeks after the injections, mice were bilaterally implanted with custom-built optic fibers (diameter: 200 μm ; numerical aperture: 0.39; Thorlabs) above the vIPAG at the following coordinates relative to bregma: i) Sst-IRES-Cre mice: -4.4 mm AP ; $\pm 1.0\text{ mm ML}$; -1.8 mm DV from dura, with a 10 degrees angle; ii) VIP-IRES-Cre mice: -4.4 mm AP ; $\pm 0.8\text{ mm ML}$; -2.0 mm DV from dura, with a 10 degrees angle. For RVM manipulations, mice were implanted at the following coordinates relative to bregma: -5.8 mm AP ; 0.0 mm ML ; -5.2 mm DV from the dura. All implants were secured using three stainless steel screws and Super-Bond cement (Sun Medical). During surgery, long- and short-lasting analgesic agents were injected (Metacam, Boehringer; Lurocaïne, Vetoquinol). After surgery, mice were allowed to recover for at least five days. Afterward, mice were handled daily to familiarize themselves with being restrained for the optic fibers connection. Behavioral experiments were performed at least four weeks after viral injections. Only mice with correct placement of optic fibers and virus expression were included in the analyses.

For optogenetic excitation, light stimulation consisted of blue light (473 nm, $\sim 8\text{--}10\text{ mW}$ at fiber tip) delivered with 2 Hz frequency and 5 ms pulse duration. In contrast, light stimulation consisted of green light (532 nm, $\sim 8\text{--}10\text{ mW}$ at fiber tip) delivered continuously for optogenetic inhibition. For optogenetic manipulations during the CFCA paradigm of either SST⁺ or VIP⁺ vIPAG cells, the light was delivered during the FC HP test and paired with the tone presentation.

Two sets of experiments were performed sequentially for stimulation during the fear conditioning. First, the footshock US was replaced by optical stimulation. Then, the US became the optical stimulation combined with the footshock (Supplementary Figure 6). For both conditions, the optogenetic stimulation started 5 s before and lasted until 5 s after CS⁺ offset. The same mice were used for both experiments. For the manipulations during the fear retrieval, there were 12 CS⁺ presentations divided into blocks of 4 CS⁺. The optogenetic stimulation was paired with the second block of CS⁺.

Anatomical tracing

Fluorogold injection

An incision between one to two cm was made slightly caudal to the peak of the dorsal hump to expose the lumbar spinal region. The vertebra of interest was identified. Then, a small incision was made between the tendons and the vertebral column on either side. The vertebra was then secured using spinal adaptor clamps, and all tissue was removed from the surface of the bone. Pulled borosilicate glass capillaries (Ringcaps, disposable capillary pipettes with ring mark, DURAN, Hirschmann Laborgeräte, Germany) were inserted between 2 vertebrae and allowed to microinject 0.5 μL of fluorogold 2% in the dorsal horn of the spinal cord on both sides.

In vivo electrophysiology

Mice were anesthetized with isoflurane 4% for induction, then 1.5% maintenance. The experiment was started as soon as there was no longer any reflex. The colorectal temperature was kept at 37 °C with a heating blanket. Two metal clamps were used to set the animal spine in a stereotactic frame (M2E, France) for stability during electrophysiological recordings. Then, a laminectomy was performed at T13-L1 to expose the lumbar part of the spinal cord. The dura mater was carefully removed. Custom-made optical fibers were placed 1mm above the dorsal spinal cord for optogenetic manipulations. C-fiber-evoked field potentials were recorded in the deep lamina of the DH (at a depth range of 250 and 500 μm) with borosilicate glass capillaries (2 M Ω , filled with NaCl 684 mM; Harvard Apparatus, Cambridge, MA, USA). Field potentials were recorded with an ISODAM-amplifier (low filter: 0.1Hz to high filter: 0.1 kHz; World Precision Instruments, USA) in response to electrical stimulation of the ipsilateral paw. Single unit recordings of WDR DH cells were made with the same borosilicate glass capillaries mentioned above and placed in the dorsal

part of the spinal cord. The criterion for selecting a cell was the presence of an A-fiber-evoked response (0-80 ms) followed by a C-fiber-evoked response (80-150 ms) to electrical stimulation of the ipsilateral sciatic nerve.

The paw of the mice was stimulated by trains (every 30 s) of electrical stimulation at two times the threshold for C-fibers were performed before, during, and after optogenetic stimulations with an optic fiber placed above the recording site. Subthreshold stimulations were performed below the threshold for C-fiber and A-fiber, respectively.

Histology analyses

Mice were administered a lethal dose of Exagon and underwent transcardial perfusions via the left ventricle with 4% w/v paraformaldehyde (PFA) in 0.1 M PB. Following dissection, brains were post-fixed for 24 h at 4°C in 4% PFA. Brain sections of 80 µm-thick were cut on a vibratome, mounted on gelatin-coated microscope slides, and dried. To verify correct viral injections and optic fiber location, serial 80 µm-thick slices containing the regions of interest were mounted in VectaShield (Vector Laboratories) and imaged using an epifluorescence system (Leica DM 5000) fitted with a 10-x dry objective. The location and the extent of the injections/infections were visually controlled. Only infections targeting the vPAG and optic fibers terminating, depending on the experiment, above the vPAG or RVM were included in the analyses.

Statistics

All statistics details are contained in Supplementary Table 1. Box-whisker plots indicate median, interquartile range, and 5th - 95th percentiles of the distribution. All statistics are indicated where used. Statistical analyses were performed with Prism software (GraphPad). We tested the normality of all data with the Kolmogorov–Smirnov normality test in Prism software (GraphPad), and used nonparametric statistical tests for nonnormally distributed datasets, as indicated where used. No statistical methods were used to predetermine sample sizes, but sample sizes were based on previous studies in our laboratory. Blinding for the opsin was done for optogenetics experiments. Animals used for the CFCA paradigm that did not satisfy the DI and CI criteria after two conditioning sessions were discarded from the study. Mice with an incorrect injection of the opsins or misplaced location of the optic fibers were discarded. For *in vivo* electrophysiology, field potentials were measured as the area above the curve in the C-fiber range (80-300 ms). Absolute values were used to compare values in each condition (OFF vs. ON optogenetic manipulation). In single-unit recordings, the number of A- and C-fiber induced spikes of WDR cells were measured after each electrical stimulation, during, and after optogenetic manipulation. An average of the four stimuli for each WDR recorded was used for statistical analysis. Outliers were removed using the Tukey method. No other mice were excluded. Significance levels are indicated as follows:

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

Data and script availability

Data supporting the findings of this study and data analysis codes are available from the corresponding author on request.

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Additional information

Author Contributions

N.W performed behavioral experiments and optogenetic experiments on freely moving animals. F.A, J.V, performed electrophysiological and optogenetic experiments on anesthetized animals. E.V, L.C, N.W, F.A, Z.G, T.B, C.R, R.B, and D.G performed histology. N.W, M.L, S.V. P.F, and C.H. designed the experiments. N.W, F.A, D.J, and E.V analyzed the data, N.W, D.J. and C.H wrote the paper.

Supplementary Figures

Supplementary Figure 1. Cued fear-conditioned analgesia behavior. **a.** During habituation, the freezing levels for the context (BL) and the two tones were low but significantly different (****, $P = 0.0003$, Friedman test, $n = 12$ mice). **b.** Mean temperature at which the NC response was observed in the HP test. The temperature of NC response was higher during CS^+ trials when compared to the CS^- trials (***, $P = 0.0005$, Wilcoxon matched-pairs signed rank test, $n = 12$ mice). **c.** Schematic representation of how the back and tail temperature of the mice was measured. Every 30-sec three-point temperature was taken for each of the two body parts. The temperature of the mice back and tail were measured by the infrared digital thermographic camera and analyzed offline (see Methods). The average temperature of the mice back (**d**) and tail (**e**) while the CS^+ or the CS^- were presented. There were no differences in body temperature for the different CSs trials (ns, $P > 0.05$, mixed-effects model, $n = 13$ mice). Vertical dashed lines correspond to the average time of NC response for the CS^+ (red) and CS^- (blue) during the standard CFCA protocol.

Supplementary Figure 2. CFCA depends on associative processes. **a. Top**, protocol for the *No Tone*: naïve mice were submitted to two HP trials without conditioning nor tone presentation. **Bottom**, protocol for the *Tone*: naïve mice were submitted to two HP trials paired with an unconditioned tone presentation. **b, c.** Mean latency and temperature of NC response for the two tests mentioned above and the CFCA. The transient cued-fear induced analgesia (CS^+) compared to the basal nociception, the tone, and the CS^- trials (***, $P = 0.0001$, Krustal-Wallis test). Trials between the same type of test (*No tone* or *Tone*) were not significantly different (ns, $P > 0.05$, Bonferroni post-hoc test, see statistical table for all comparisons) **d.** After fear extinction, there was no difference in the mean temperature response between the two CSs (ns, $P = 0.3276$ paired t-test, $n = 10$ mice). **e.** Protocol for stability training (see methods). Mice were submitted to two rounds of the CFCA paradigm. During retrieval (**f, g**), average freezing values during CS^+ was higher than CS^- or baseline (BL) periods (****, $P < 0.0001$, one-way RM ANOVA, $n = 10$ mice). **h, i.** Mean latency and temperature of NC response during CS^- and CS^+ trials for FC HP test 1 (**h**) and test 2 (**i**). CS^+ presentation increased the latency and temperature of NC response in FC HP test 1 and test 2 (*, **, and ***, $P < 0.05$, paired t-test; for details, see supplementary table 1).

Supplementary Figure 3. Two distinct neuronal populations in SST-Cre mice within the PAG. **a.** Representative picture of single-molecular fluorescent in situ hybridization for Sst mRNAs in the PAG. Scale bar, 400 μm . **b, c.** Single-molecular fluorescent in situ hybridization for Sst (green) and Cre (red) mRNAs in the vIPAG (left). Histograms showing the co-expression of Sst/Cre as percentage of Sst-expressing cells (green) and as percentage of Cre-expressing cells (red) in the dIPAG and vIPAG (right). Scale bar, 20 μm . **d.** Single-molecular fluorescent in situ hybridization for Sst (red) and Slc32a1 (green) within the dIPAG (upper panel) and vIPAG (bottom panel). Scale bar, 20 μm . **e.** Quantification of colocalization within the dIPAG (upper panel) and vIPAG (bottom panel) of Sst^+ and $Slc32a1^+$. In the dIPAG approximately 19% of $Slc32a1^+$ cells are Sst^+ and 17% of Sst^+ cells are $Slc32a1^+$. On the contrary, in the vIPAG approximately 95% of $Slc32a1^+$ cells are Sst^+ and 61% of

Sst⁺ cells are Slc32a1⁺. **f.** Single-molecular fluorescent in situ hybridization for Sst (red) and Slc17a6 (green) within the dIPAG (upper panel) and vIPAG (bottom panel). Scale bar, 20 μ m. **g.** Quantification of colocalization within the dIPAG (upper panel) and vIPAG (bottom panel) of Sst⁺ and Slc17a6⁺. In the dIPAG approximately 91% of Slc17a6⁺ cells are Sst⁺ and 38% of Sst⁺ cells are Slc17a6⁺. On the contrary, in the vIPAG approximately 34% of Slc17a6⁺ cells are Sst⁺ and 33% of Sst⁺ cells are Slc17a6⁺. White arrows indicate colocalization.

Supplementary Figure 4. Comparable fear levels prior to the FC HP test. After retrieval, fear conditioning levels between the opsins and their respective GFP group was tested to ensure equivalent fear levels. The discrimination index (see methods) between the GFP and Chr2 (a, ns, $P = 0.6052$ unpaired t-test, $n = 16$ mice) or ArchT (b, ns, $P = 0.4575$, unpaired t-test, $n = 23$ mice) were not significantly different. All the mice discriminated equally between CSs. The conditioning index (see methods) between the GFP and Chr2 (c, ns, $P = 0.2333$, unpaired t-test $n = 16$ mice) or ArchT (d, ns, $P = 0.1320$, unpaired t-test, $n = 23$ mice) were also not significantly different. All mice displayed a similar high freezing level to the CS⁺.

Supplementary Figure 5. Optogenetic manipulation of SST⁺ vIPAG cells does not affect locomotion or produce aversion. **a.** Photoactivation of SST⁺ vIPAG cells abolished the analgesic effect induced during defensive states (**, $P = 0.0035$, opsin x CS - two-way RM ANOVA, $n = 7$ GFP, $n = 9$ Chr2). The mean temperature of NC response for the CS⁺ was significantly different between the Chr2 and GFP groups (*, $P = 0.0322$, Bonferroni post-hoc test). For the Chr2 group, the mean temperature of NC response during CS⁺ was equivalent to the CS⁻ (ns, $P = 0.1783$, Bonferroni post-hoc test). On the contrary, the mean temperature of NC response between the CSs differed for the GFP group (****, $P < 0.0001$, Bonferroni post-hoc test). **b.** Photoinhibition of SST⁺ vIPAG cells increased the analgesic effect for the ArchT group when compared to the GFP (**, $P = 0.0035$, opsin x CS - two-way RM ANOVA, $n = 12$ GFP, $n = 11$ ArchT). The mean temperature of NC response for the CS⁻ was significantly different between the ArchT and GFP groups (****, $P < 0.0001$, Bonferroni post-hoc test). For the GFP group, the mean temperature of NC response was higher for the CS⁺ trials when compared to the CS⁻ trials (***, $P = 0.0003$, Bonferroni post-hoc test), yet this was not the case for the ArchT group (ns, $P > 0.999$, Bonferroni post-hoc test). **c.** The photoactivation of SST⁺ vIPAG cells was performed during the ON epoch (blue shaded area), and the average distance travelled during the ON epoch was not different from the OFF epochs when comparing the two opsins (ns, $P = 0.5121$, opsin x light - two-way RM ANOVA, $n = 7$ GFP, $n = 9$ Chr2). **d.** The photoinhibition of SST⁺ vIPAG cells was performed during the ON epoch (yellow shaded area), and the average distance travelled during the ON epoch was not different from the OFF epochs when comparing the two opsins (ns, $P = 0.4047$, opsin x light - two-way RM ANOVA, $n = 12$ GFP, $n = 11$ ArchT). **e.** Real-time place-preference location plot from a representative animal while submitted to optogenetic activation of SST⁺ vIPAG cells in the left compartment throughout the 15-min session. **f.** There was no difference between Chr2-expressing mice and the control group in the time spent on the stimulated and non-stimulated compartments (ns, $P = 0.1812$, Mann Whitney test).

Supplementary Figure 6. SST⁺ vIPAG cells photoactivation during conditioning does not impact fear learning. **a.** Protocol for optogenetic manipulations during fear conditioning. On Day 2, mice received 5 CS-US associations. The US was either optogenetic stimulation alone or optogenetic stimulation plus footshock. **b.** Average freezing levels during the CS-US association of optogenetic stimulation alone. There was no difference in the overall freezing levels for the CS⁻ and CS⁺ between Chr2 and GFP (ns, $P = 0.4804$, opsin x CS - two-way RM ANOVA, $n = 6$ GFP, $n = 7$ Chr2). **c.** During retrieval, there was no effect on freezing upon activation of SST⁺ vIPAG cells as an US (ns, $P = 0.4804$, opsin x CS effect - two-way RM ANOVA, $n = 6$ GFP, $n = 7$ Chr2). **d.** Average freezing levels during the CS-US association of optogenetic stimulation plus footshock. There was no difference in the overall freezing levels for the CS⁺ and CS⁻ between Chr2 and GFP (ns, $P =$

0.8663, opsin x CS effect - two-way RM ANOVA, $n = 6$ GFP, $n = 7$ Chr2). e. During retrieval, there was no difference in the fear expression by the activation of the SST⁺ vIPAG cells (ns, $P = 0.8967$, opsin x CS effect - two-way RM ANOVA, $n = 6$ GFP, $n = 7$ Chr2).

Supplementary Figure 7. Optogenetic effect observed during CFCA is not due to alteration of somatostatin levels nor mediated by VIP⁺ vIPAG cells.

a. Sst-IRES-Cre heterozygotic mice received a bilateral injection of opsins in the vIPAG, and optic fibers were implanted above the vIPAG. b. A representative example of expression patterns of Chr2 within SST⁺ vIPAG cells. c. Average freezing values during retrieval. Chr2 and GFP groups were pooled together because no differences were found in the conditioning level (data not shown). The average freezing values during CS⁺ was higher than CS⁻ or baseline (BL) periods (****, $P < 0.0001$, Friedman test, $n = 13$ mice). d. Optogenetic activation of SST⁺ vIPAG cells abolished the analgesic effect induced by defensive states (**, $P = 0.0048$, opsin x CS - two-way RM ANOVA, $n = 6$ GFP, $n = 7$ Chr2). The mean latency of NC response for the CS⁺ was significantly different between the Chr2 and GFP groups (*, $P = 0.0300$, Bonferroni post-hoc test, $n = 6$ GFP, $n = 7$ Chr2). For the Chr2 group, the latency of NC response during CS⁺ was equivalent to the CS⁻ (ns, $P = 0.3342$, Bonferroni post-hoc test). On the contrary, the latency of NC response between the CSs was different for the GFP group (****, $P = 0.0001$, Bonferroni post-hoc test). e. VIP-IRES-Cre mice received a bilateral injection of opsins in the vIPAG, and optic fibers were implanted above the vIPAG. f. Representative example of the expression pattern of ArchT within VIP⁺ vIPAG cells. g. ArchT and GFP groups were pooled together because no differences were found in the conditioning level (data not shown). The average freezing values during CS⁺ was higher than CS⁻ or BL periods (****, $P < 0.0001$, one-way RM ANOVA, $n = 10$ mice). h. Optogenetic inhibition of VIP⁺ cells did not change the analgesic effect for the ArchT group when compared to the GFP (ns, $P = 0.5455$, opsin x CS effect - two-way RM ANOVA, $n = 5$ GFP, $n = 5$ ArchT). The latency of NC response was higher for the CS⁺ trials compared to the CS⁻ trials for both GFP and ArchT (***, $P < 0.01$, Bonferroni post-hoc test, $n = 5$ GFP, $n = 5$ ArchT).

Supplementary Figure 8. Optogenetic activation of SST⁺ vIPAG cells did not affect tactile sensitivity.

a. Perievent stimulus time histogram indicating the spiking latency of a purely tactile neuron recorded in the spinal cord in response to gentle brushing of the skin. b. Representative spiking activity latency of a purely tactile neuron recorded in the spinal cord in response to gentle brushing of the skin with or without activation of Chr2 expressing SST⁺ vIPAG cells. c. The activation of Chr2 expressing SST⁺ vIPAG cells of tactile neuron recorded in the spinal cord in response to gentle brushing of the skin had no effect on the average firing activity (Paired t-test, $P = 0.6929$, $n = 5$ cells).

Supplementary Figure 9. Optogenetic manipulation of GFP expressing vIPAG cells did not affect analgesia.

a. Single-unit recordings of WDR cells in the lumbar spinal cord during optogenetic manipulation of GFP expressing SST⁺ vIPAG inputs to the RVM. b. Photostimulation of SST⁺ vIPAG inputs to the RVM has no effect.

Additional files

Supplementary Figures [↗](#)

Supplementary Table [↗](#)

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Reviewer #1 (Public review):

Summary:

In the manuscript by Winke et al, the authors present evidence that fear-induced analgesia is mediated by somatostatin projection cells from the vlPAG to the RVM. This study uses a mouse model of fear-induced analgesia, and incorporates optogenetic circuit manipulation with behaviour and electrophysiology to gain a meaningful insight into a novel circuit involved in fear-induced analgesia.

Strengths:

- (1) This is a well-constructed study with appropriate controls and analyses.
- (2) Alternative interpretations of the data are systematically considered and eliminated via rational experiments. The authors are commended for a nice piece of experimental work.
- (3) The vlPAG is a known region of pain modulation, and this study adds valuable insight to the circuit involved in fear-associated analgesia.

Weaknesses:

- (1) Only male mice are included in this study.
- (2) Animals are excluded from analyses based on clearly defined criteria, but it is not clear how many mice were excluded from each group.
- (3) The authors implement a pain sensitivity assay that involves a hot plate with progressively increasing temperature. The time to nociceptive responses is reported. Without reporting the actual temperature at which the mice respond, it makes it difficult to compare nociceptive responses to previously published work (which typically use a defined and static hotplate temperature).

(4) The authors present evidence that inhibition of SST vIPAG cells enhances spinal nociceptive electrophysiological responses, but the corresponding pain sensitivity is not altered (Figure 2, CS- condition). The reason for the discrepancy between electrophysiological and behavioural responses is not clear.

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Reviewer #2 (Public review):

Summary:

Wenke et al. investigated the role of vIPAG somatostatin-expressing neurons in the mediation of analgesia during defensive states. A newly developed paradigm of cued fear-conditioned analgesia, which consists of a combination of an auditory fear retrieval session and a pain test, was used to evaluate this cell population's contribution to fear-mediated analgesia. Optogenetic manipulation of vIPAG SST+ neurons modulated the responses to a nociceptive cue (Hot Plate) presented concomitantly with an aversively conditioned tone. At the same time, alterations in the freezing levels could be observed during optogenetic activation of vIPAG SST+ neurons. In order to disentangle the impact of these cells on analgesia from their impact on the expression of defensive behaviors, the authors performed electrophysiological recordings from the dorsal horn in the spinal cord of anesthetized mice. A vIPAG-RVM-DH pathway was identified to trigger nociceptive C-fibers upon optic activation of the RVM. Finally, pathway-specific activation of SST+ vIPAG-RVM neurons could abolish CS-induced analgesia.

Strengths:

The study addresses a relevant topic, that is, brainstem circuits for pain-modulatory mechanisms as part of defensive states evoked by threat. This is important because the circuit mechanisms underlying pain are still not fully understood, and defining molecular markers of cellular circuit substrates may support the identification of potential pharmaceutical targets in treating pain. The authors confirm a previous study in that a somatostatin-positive cellular population presents a crucial vIPAG circuit element mediating anti-nociceptive effects. Key novelty aspects of the present study are the demonstration that these neurons seem to play a role specifically in threat-induced analgesia. This was possible by the elegant design and application of a novel fear analgesia paradigm, combined with cell- and pathway-specific optogenetics.

Weaknesses:

Despite the convincing and rigorous experimental approach, the study leaves some interpretational room when it comes to the proposed circuit mechanism. This could either be addressed by additional experiments or by more discussion of alternative circuit layouts.

Major Comments:

(1) The paper by Zhang et al. (<https://pubmed.ncbi.nlm.nih.gov/36641028/>), which identified a role for vIPAG SOM+ neurons in mediating anti-nociception in neuropathic pain, needs to be referenced and its results discussed, if not reconciled. While functionally, both studies find an analgetic role of vIPAG SOM+ neurons projecting to the RVM, Zhang et al., using slice physiology, characterize those neurons as glutamatergic. In Figure 4E of Zhang et al. they find general (fear-independent) analgetic effects with PAG-RVM specificity by performing chemogenetic experiments.

It can be argued that in addition to the two functionally distinct inhibitory SOM subtypes hypothesized by Winke et al., there is another, excitatory subpopulation. Also, the different experimental conditions (chronic vs. acute pain, non-threat vs. fearful cues/contexts) may recruit different vIPAG SOM+ populations. All of this is conceivable, yet I wonder whether the contrasting findings could more parsimoniously be reconciled. The author's own results presented here in Supplementary Figure 3 suggests that SOM+ vIPAG cells are co-localizing with glutamate and thus could also be excitatory. In addition to this rather complementary piece of evidence, a more extensive characterization of vIPAG neurons using IHC and slice physiology would be needed to justify the unambiguous identification of their inhibitory nature.

In the absence of a direct identification of these cells exclusively releasing GABA, an alternative explanation should be considered. What about looking at vIPAG SOM+ neurons as a putatively mixed bag of local, inhibitory interneurons and long-range, RVM-projecting excitatory cells? This model would then open up interesting questions as to the actual function of somatostatin as a modulator of vIPAG circuit activity and associated function, and from my perspective, would nicely fit into the view of PAG circuits as integrators of complex survival responses.

(2) "Our data indicate that the optogenetic inhibition of SST+ vIPAG cells promotes analgesia irrespective of the animal's defensive state. In contrast, the optogenetic activation of long-range SST+ vIPAG cells that project to the rostral ventromedial medulla (RVM) abolishes the analgesia mediated by fear behavior." (lines 32-35). Consider toning down these conclusions, as contrasting activation with inhibition of two different (though overlapping) populations cannot be fully conclusive. Alternatively, a pathway-specific (vIPAG-RVM) inhibitory experiment could help to fully understand the circuit mechanism and verify the necessity of these neurons.

(3) Despite an overall very thorough reporting style, some information is missing from the manuscript:

a) In Figures 2d and f, what are the freezing levels during optogenetic manipulation? From Figure 3d, one can expect that freezing is inhibited during the hot plate test, which could bias the NC response towards shorter latencies. b) In Figure 5, the histological experiment showing the vIPAG-to-RVM pathway is presented by a qualitative image only. Here, some quantification would strengthen the finding. c) In Figures 6 c and d "Consistently, activation of the SST+ vIPAG-RVM pathway during CFCA had no impact on CS-presentation, whereas the same manipulation performed during CS+ blocked the increase in NC response latency compared to GFP controls." (line 194-196). Is it possible that the NC response cannot be any lower than the one during CS-, thus constituting a floor effect? d) Connected to major point 1- this experiment is important for defining the circuit mode and therefore should be as convincing as possible. However, for the colocalization experiment in Supplementary Figure 3, the methodological description is missing and thus makes it hard to comprehend how this data set was generated (how many data points, etc.). The visual depiction of the results is non-standard and not easily graspable. Consider e.g., a Venn diagram.

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Reviewer #3 (Public review):

Summary:

Conditioned analgesia refers to the ability of a learned fear cue to suppress pain-related behavior and neural activity. Understudied, the authors developed a novel conditioned

analgesia procedure in which a cue that had been paired or unpaired with shock was played while a hot plate increased temperature. Compared to several control conditions, the authors found increased latency to a nociceptive response (paw licking). The authors identified somatostatin neurons in the periaqueductal gray as a likely mediator of the behavior. They then showed that: (1) stimulating vlPAG-SST neurons blocked nociceptive response latency increases to the CS+, (2) stimulating vlPAG-SST neurons suppressed fear retrieval freezing, (3) stimulating vs. inhibiting vlPAG-SST neurons drove opposing modulation of c-fibers and Aδ-fibers, (4) direct-projecting vlPAG SST neurons modulate freezing while RVM-projecting vlPAG SST neurons modulate conditioned analgesia.

Strengths:

These experiments have many strengths. The behavioral assay is chief among them. The assay is robust and controls for confounding factors to reveal a repeatable effect of a shock-paired cue to delay nociceptive responding. The optogenetic experiments provide the correct level of temporal precision, given the authors' time-specific interest in cued responding. Combining neuronal manipulations with spinal recordings is particularly innovative, especially in the context of more behavioral neuroscience-based assays. All-in-all, I found this to be an exceptionally strong set of experiments.

Weaknesses:

No obvious weaknesses were identified by this Reviewer.

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